



6.3.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...



Unit 6, Lesson 4: Sequence of Transformations



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



Learning Targets

- ☐ I can represent a sequence of transformations that will map the preimage onto the image.
- ☐ I can describe a sequence of transformations that will map the preimage onto the image.
- ☐ I can identify transformations that do or do not preserve distance.
- ☐ I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.



6.4.1 Warm-Up: Would You Rather?

1. Would you rather drive a car at a rate of 40 miles per hour, or drive a car at a rate of 50 feet per second? Remember: There are 5280 feet in a mile.

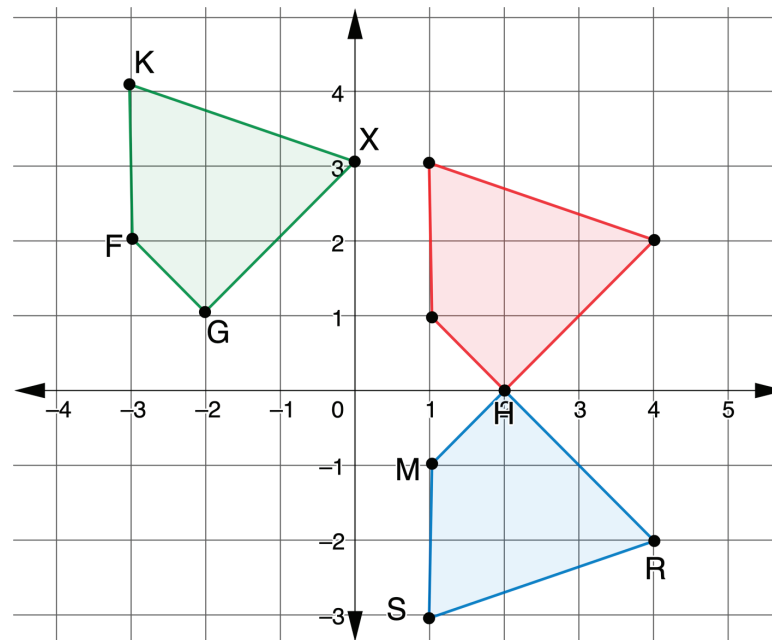


Explain your choice using mathematics.



6.4.2 Guided Instruction: Sequence of Transformations

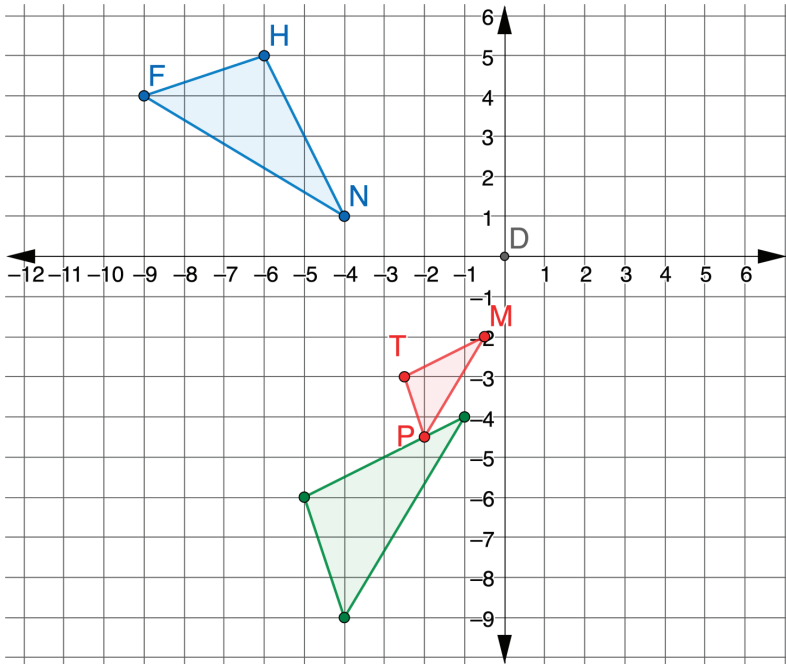
1. $XGFK$ is the result of a sequence of transformations of $RSMH$. The red quadrilateral is the first transformation.



- a. Describe the first transformation of $RSMH$.

- b. Write the transformation algebraically.
- c. Describe the second transformation of $RSMH$ that results in $XGFK$.
- d. Write the transformation algebraically.
- e. Explain if the sequence of transformations preserves distance.

2. $\triangle MTP$ is the result of a sequence of transformations of $\triangle HNF$.

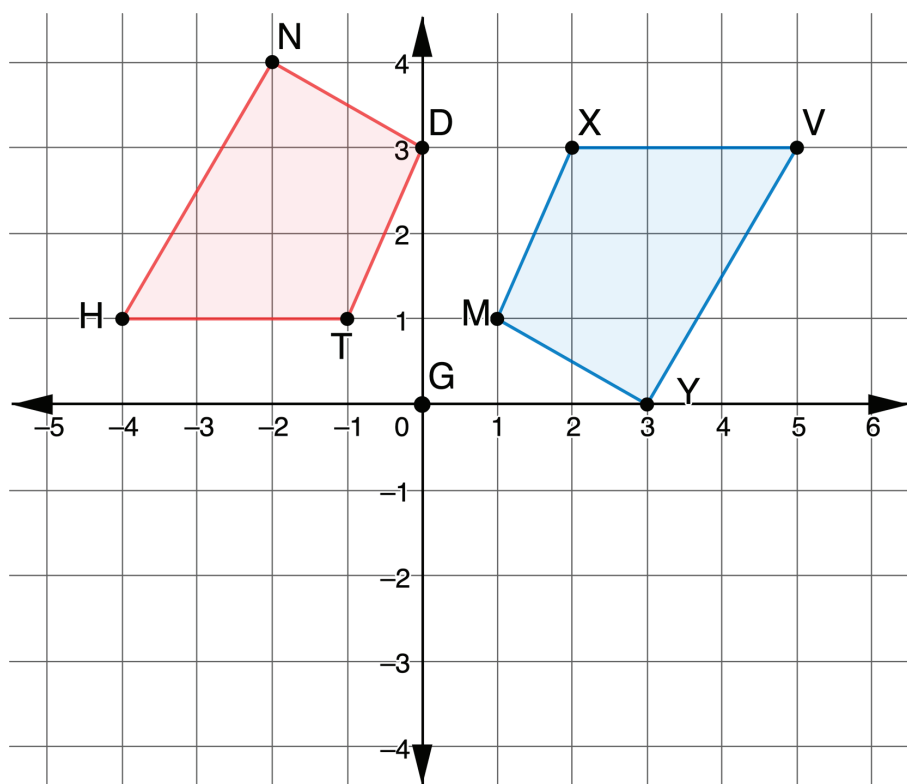


a. Determine the sequence of transformations that will map $\triangle HNF$ onto $\triangle MTP$. Complete the table.

	$\triangle HNF$ Transformed to $\triangle H'N'F'$	$\triangle H'N'F'$ Transformed to $\triangle MTP$
Written Description		
Algebraic Description		

- b. Explain how to determine if this sequence of transformations preserves distance.

3. $TDNH$ is the result of a sequence of transformations of XYV .



- a. Determine a sequence of transformations that will map XYV onto $TDNH$. Complete the table.

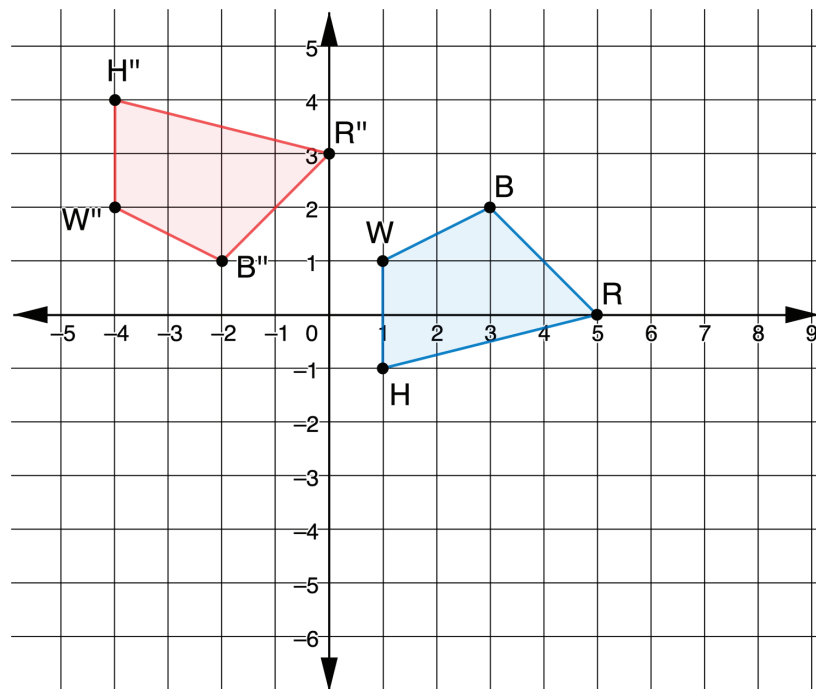
	First Transformation	Second Transformation
Written Description		
Algebraic Description		

- b. Which transformations preserve distance?



6.4.3 Guided Practice: Mapping a Given Preimage onto an Image

$WBRH$ was transformed by a sequence of transformations to create $W''B''R''H''$.



Esmeralda and Wesley were determining the sequence of transformations and wrote their answers in the table.

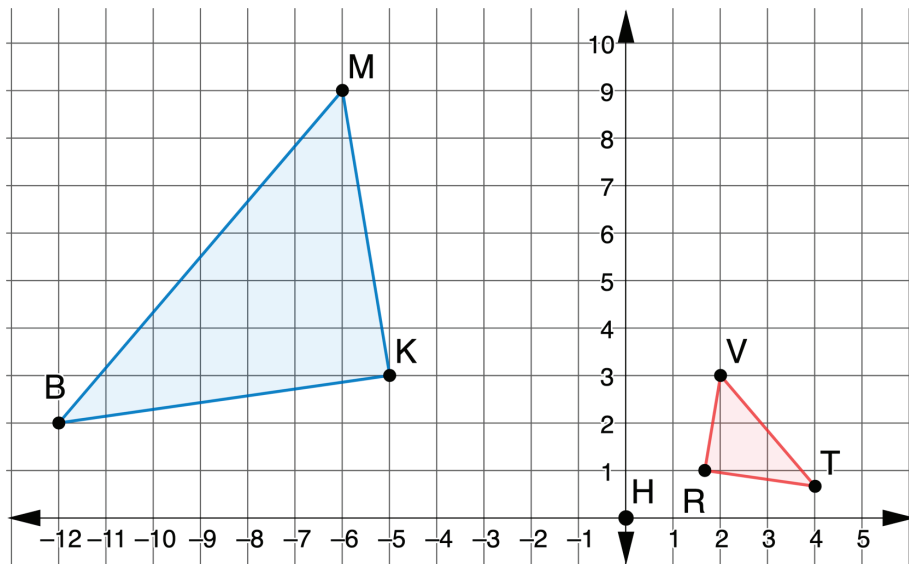
	$WBRH$ Transformed to $W'B'R'H'$	$W'B'R'H'$ Transformed to $W''B''R''H''$
Esmeralda's Answer	Translated 5 units to the left and 3 units down	Reflected over the x -axis
Wesley's Answer	Translated 3 units down	Rotated 180° about the origin

1. Determine whose sequence of transformations is correct. Explain your reasoning.



6.4.4 Your Turn!

1. $\triangle MKB$ was transformed to create $\triangle VRT$.

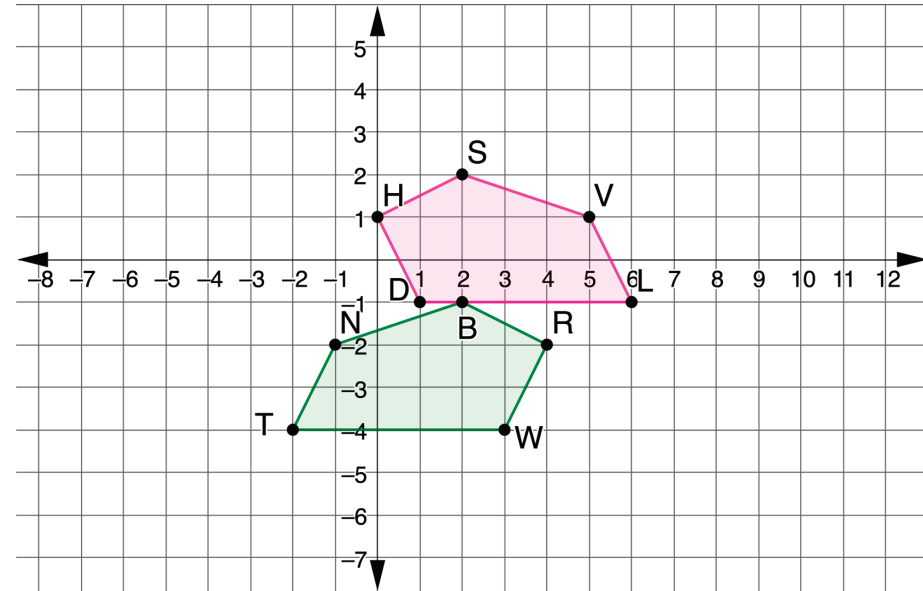


- Describe a sequence of transformations that will map $\triangle MKB$ onto $\triangle VRT$.
- Write the sequence of transformations using algebraic descriptions.



6.4.5 Wrap-Up: Ticket Out the Door

1. A sequence of transformations can map $NBRWT$ onto $VSHDL$.



Determine a sequence of transformations that will map $NBRWT$ onto $VSHDL$. Complete the table.

	First Transformation	Second Transformation
Written Description		
Algebraic Description		